



Air quality monitoring at the Cemex plant in Monterrey

CLIENT

Cemex

SECTOR

Industry

LOCATION

Monterrey, Mexico

PRODUCTS

Kunak AIR Pro
Kunak AIR Cloud

Need

Cemex, one of the world's leading cement manufacturers develops a proactive policy that seeks to resolve in advance the possible problems that its production activity may generate. For this reason, it asked us to deploy a solution that would protect the health of the population and workers at its facilities, as well as ensure the conservation of the natural environment.

To this end, they contacted Rossbach, the distributor of Kunak products in Mexico, to ask for the installation of a perimeter monitoring ring at its production plant in Monterrey (Mexico).

This Mexican multinational has been investing heavily in sustainability for several years, so this measure was a continuation of its commitment to the well-being of the local population, its employees and compliance with environmental regulations.

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Solution

Deployment of a perimeter network of Kunak AIR Pro stations for air quality monitoring as well as an anemometer and rain gauge.

With this network different parameters are measured continuously thus providing information on the AQI (Air Quality Index).

Measured parameters:

- PM₁₀, PM_{2.5} and PM₁, Total Suspended Particulate Matter (TSP) and Total Particulate Counter (TPC).
- Sulphur oxide (SO₂), nitrogen dioxide (NO₂), ozone (O₃) and carbon monoxide (CO).
- Temperature, relative humidity and atmospheric pressure.
- Wind speed and direction.
- Collection and measurement of precipitation.

Benefits

- Emission reductions through monitoring and accurate knowledge of emissions.
- Achievement of an excellent quality of data on the main pollutants arising from cement production.
- Protection of the health of people living near the plant and of the employees.
- Compliance with environmental regulations and standards set by local and national authorities.
- Creation of an early warning system that is activated when any of the measured parameters exceeds the established threshold values.

Results

Deployment of an environmental monitoring network that provides excellent data quality on the main pollutants derived from cement production and design of an early warning system when any parameter exceeds the established limit values to thus ensure and protect the health of both plant workers and nearby populations.

In 2019, CEMEX achieved a large reduction in its emissions compared to 2005 levels, year zero, when it began investing in improving air quality, making this multinational a guarantor of sustainability and human health.

